

21. The Cartesian product $A \times A$ has 16 elements among which are (0, 2) and (1, 3). Which of the following statements is/are correct ?

1. It is possible to determine set A.
2. $A \times A$ contains the element (3, 2).

Select the correct answer using the code given below :

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

22. Let $A = \{1, 2, 3, \dots, 20\}$. Define a relation R from A to A by $R = \{(x, y) : 4x - 3y = 1\}$, where $x, y \in A$. Which of the following statements is/are correct ?

1. The domain of R is $\{1, 4, 7, 10, 13, 16\}$.
2. The range of R is $\{1, 5, 9, 13, 17\}$.
3. The range of R is equal to codomain of R.

Select the correct answer using the code given below :

- (a) 1 only
- (b) 2 only
- (c) 1 and 2
- (d) 2 and 3

23. Consider the following statements :

1. The relation f defined by

$$f(x) = \begin{cases} x^3, & 0 \leq x \leq 2 \\ 4x, & 2 \leq x \leq 8 \end{cases} \text{ is a function.}$$

2. The relation g defined by

$$g(x) = \begin{cases} x^2, & 0 \leq x \leq 4 \\ 3x, & 4 \leq x \leq 8 \end{cases} \text{ is a function.}$$

Which of the statements given above is/are correct ?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

24. Consider the following statements :

1. $A = (A \cup B) \cup (A - B)$
2. $A \cup (B - A) = (A \cup B)$
3. $B = (A \cup B) - (A - B)$

Which of the statements given above are correct ?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

25. एक फलन $f(x-y) = \frac{f(x)}{f(y)}$ को संतुष्ट करता है,

जहाँ $f(y) \neq 0$. यदि $f(1) = 0.5$, तो

$f(2) + f(3) + f(4) + f(5) + f(6)$ किसके बराबर है ?

- (a) $\frac{15}{32}$
- (b) $\frac{17}{32}$
- (c) $\frac{29}{64}$
- (d) $\frac{31}{64}$

26. $2 \cot\left(\frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3}\right)$ किसके बराबर है ?

- (a) -1
- (b) 1
- (c) $3 + \sqrt{5}$
- (d) $3 - \sqrt{5}$

27. यदि $\sec^{-1}p - \operatorname{cosec}^{-1}q = 0$, जहाँ $p > 0, q > 0$; तो $p^{-2} + q^{-2}$ का मान क्या है ?

- (a) 1
- (b) 2
- (c) $\frac{1}{2}$
- (d) $\frac{1}{2\sqrt{2}}$

28. $1 + \sin^2\left(\cos^{-1}\left(\frac{3}{\sqrt{17}}\right)\right)$ किसके बराबर है ?

- (a) $\frac{25}{17}$
- (b) $\frac{8}{17}$
- (c) $\frac{9}{17}$
- (d) $\frac{47}{17}$

29. यदि $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$, $0 < \theta < \frac{\pi}{2}$; तो

$8 \sin^2\left(\theta + \frac{\pi}{4}\right)$ का मान क्या है ?

- (a) 16
- (b) 2
- (c) 1
- (d) $\frac{1}{2}$

30. यदि $\tan \alpha = \frac{1}{7}$, $\sin \beta = \frac{1}{\sqrt{10}}$; $0 < \alpha, \beta < \frac{\pi}{2}$, तो $\cos(\alpha + 2\beta)$ का मान क्या है ?

- (a) $-\frac{1}{2}$
- (b) $-\frac{1}{\sqrt{2}}$
- (c) $\frac{1}{\sqrt{2}}$
- (d) $\frac{1}{2}$